

**PT. INDONESIA POWER
UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM**

16.TE/CBM/UJPJRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	1 dari 8

Equipment : RAW 2 unit 1

KKS :

Technology : Vibrasi

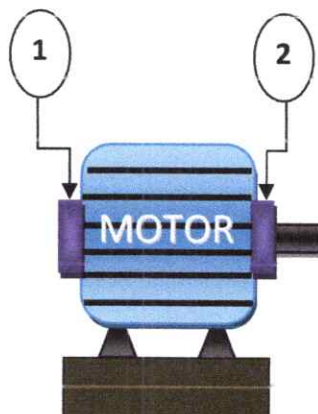
Tanggal : 13 April 2020

Status : Kuning

Finding : temperatur RAW Pump 2 unit 1 Bearing 2 Alarm

I. Data Spesifikasi Motor

MOTOR			POMPA		
Type, Mfg	:		Type, Mfg.	:	
Speed	:		Speed	:	
Power	:		Bearing Type	:	
Bearing Type	:		Inboard Bearing	:	
Inboard Bearing	:		Onboard Bearing	:	
Outboard Bearing	:		Total Blade	:	
Rotor Bar	:	-			
Foundation	:	Rigid			

FOTO EQUIPMENT	POINT PENGUKURAN
	

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INTEGRATED MANAGEMENT SYSTEM
16.TE/CBM/UJPJRJ/2020**

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	2 dari 8

II. Hasil Pengukuran Vibrasi

DATA NO LOAD SEBELUM PERBAIKAN											
TANGGAL	Durasi	VIBRASI (mm/s)						TEMPERATUR (°C)			Ket
		BEARING MOTOR						MOTOR		WINDING	
		1V	1H	1A	2V	2H	2A	1	2		
13 Mei 2020	10 menit	1.42	2.59	2.02	1.67	2.68	2.36	38.9	49.2	45.3	
	20 menit	1.26	2.41	1.9	1.74	2.46	2.31	41.6	58.1	53.1	
	30 menit	1.11	2.48	1.83	2.1	2.5	2.41	42.2	64.6	55.4	

STANDART ISO 10816-3 (GROUP 2 RIGID)				
ALARM LIMIT	≥ 0	≥ 1.4	≥ 2.8	≥ 4.5
	A	B	C	D

Hasil Rekaman Spectrum Vibrasi :

✓ Pada Motor :

- Pada Bearing 2 (DE Motor) status vibrasi Prewarning dengan nilai tertinggi 2.68 mm/s, indikasi mulai muncul kerusakan bearing

Rekomendasi :

✓ Pada Motor :

- Perencanaan penggantian bearing motor

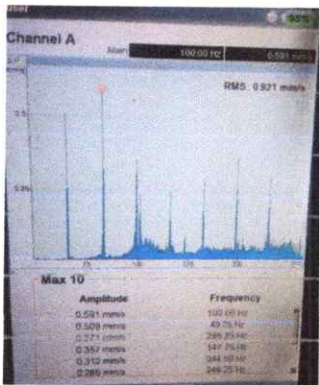
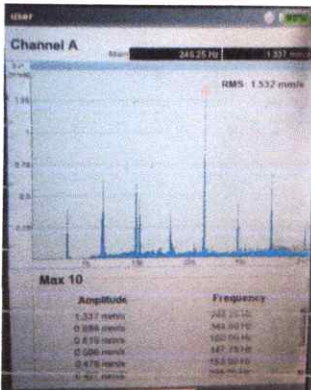
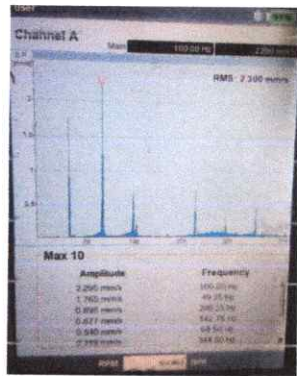


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UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM**

16.TE/CBM/UJPJRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	3 dari 8

III. Data Vibrasi :

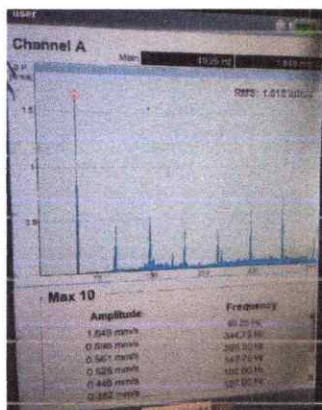
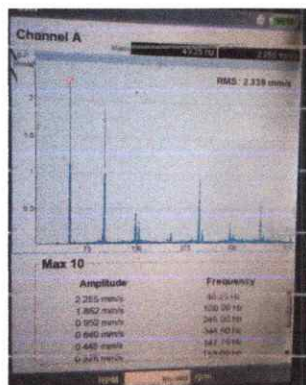
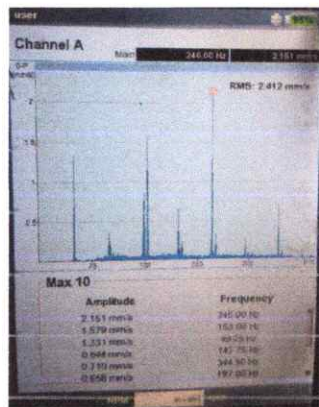
Spectrum	Keterangan														
 <table border="1"> <thead> <tr> <th>Amplitude</th> <th>Frequency</th> </tr> </thead> <tbody> <tr><td>0.521 mm/s</td><td>100.00 Hz</td></tr> <tr><td>0.509 mm/s</td><td>49.75 Hz</td></tr> <tr><td>0.271 mm/s</td><td>200.00 Hz</td></tr> <tr><td>0.357 mm/s</td><td>147.75 Hz</td></tr> <tr><td>0.312 mm/s</td><td>248.50 Hz</td></tr> <tr><td>0.285 mm/s</td><td>248.25 Hz</td></tr> </tbody> </table>	Amplitude	Frequency	0.521 mm/s	100.00 Hz	0.509 mm/s	49.75 Hz	0.271 mm/s	200.00 Hz	0.357 mm/s	147.75 Hz	0.312 mm/s	248.50 Hz	0.285 mm/s	248.25 Hz	Bearing 1V
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UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM**

16.TE/CBM/UJPRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	4 dari 8

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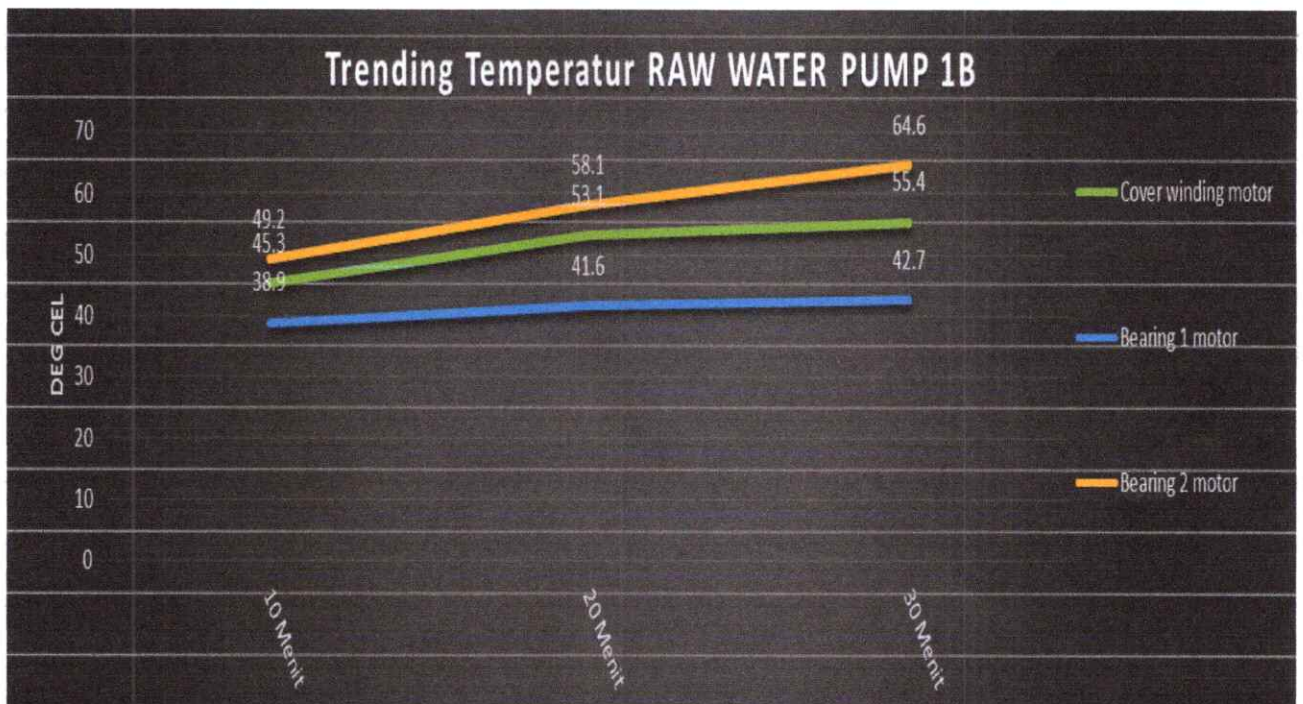
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PT. INDONESIA POWER
UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM
16.TE/CBM/UJPJRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	5 dari 8

IV. Data Thermal :

Unit	No. KKS	Description	Measurement Point	Data Analysis		
				10 Menit	20 Menit	30 Menit
1		EQUIPMENT-RAW WATER PUMP 1B	Cover winding motor	45.3	53.1	55.4
			Bearing 1 motor	38.9	41.6	42.7
			Bearing 2 motor	49.2	58.1	64.6

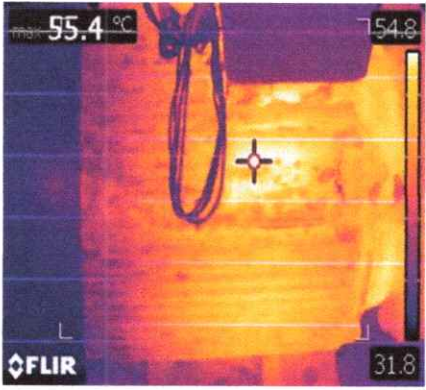
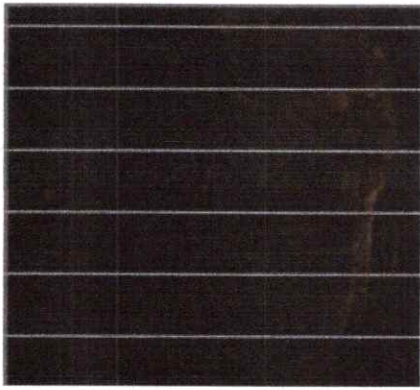
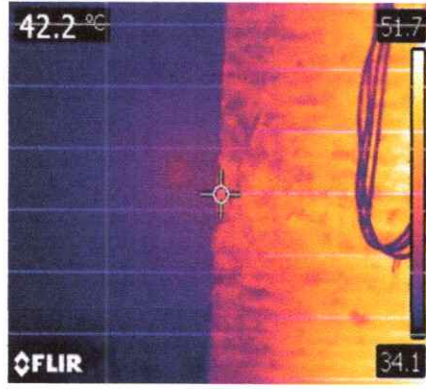
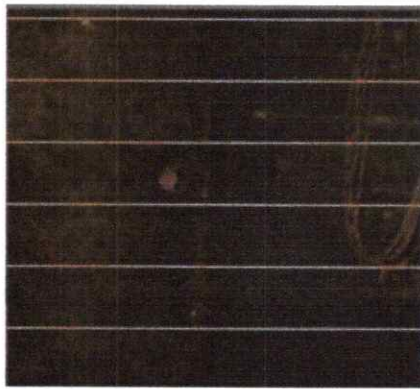
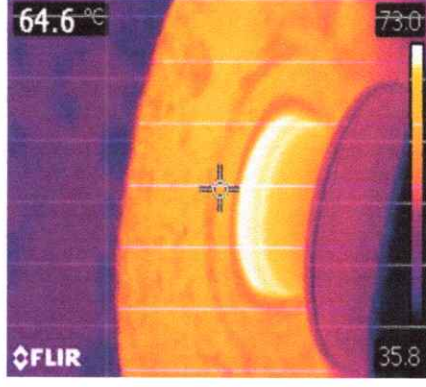
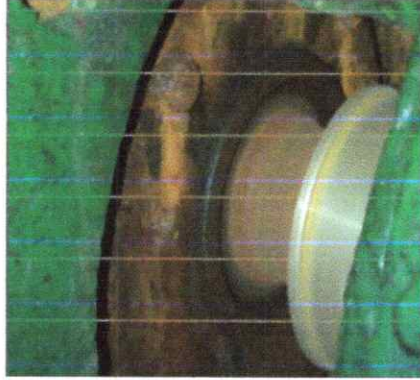


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UNIT JASA PEMBANGKITAN PLTU JERANJANG
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16.TE/CBM/UJPJRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	6 dari 8

CITRA IMAGING INFRARED THERMOGRAPHY	FOTO EQUIPMENT ACTUAL	KETERANGAN
		Pada body / cover winding RAW WATER PUMP 2 #1 temp tertinggi 55.4 degC
		Bearing 1 motor RAW WATER PUMP 2 #1 temp tertinggi 42.2 degC
		Bearing 2 motor RAW WATER PUMP 2 #1 temp tertinggi 64.6 degC

✓

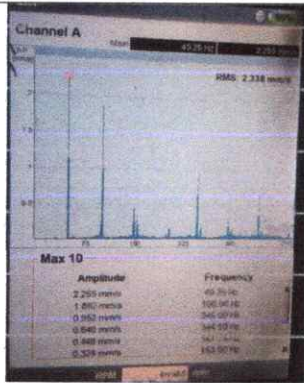
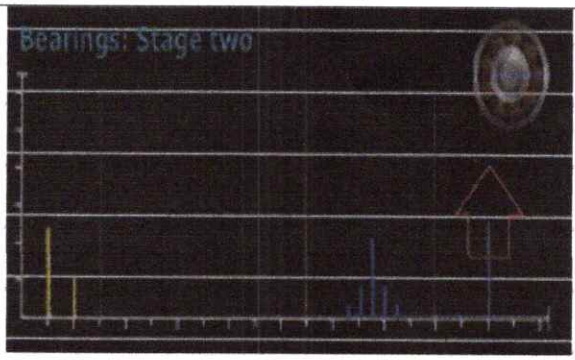
**PT. INDONESIA POWER
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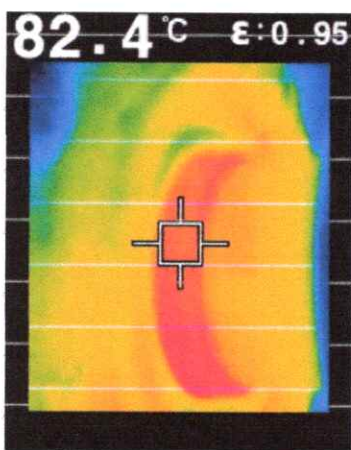
Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	7 dari 8

V. Analisa :

- Vibrasi

Syntom pengukuran	Syntom carlotte														
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Dari synton pengukuran dibandingkan dengan pola charlotte mengindikasikan terjadi kerusakan bearing stage 2															

- Thermal

CITRA IMAGING INFRARED THERMOGRAPHY	ANALISA
	Terdapat hotspot di shaft motor dengan temperature tertinggi 82.4 deg cel.

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16.TE/CBM/UJPPRJ/2020

Tgl Berlaku	No. Dokumen :	Revisi	Hal
07/02/2017	FORM.JRG.F.05.006	01	8 dari 8

IV. Kesimpulan :

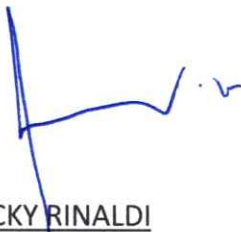
- Teknologi Vibrasi : mulai muncul indikasi kerusakan bearing stage 2
- Teknologi IRT : adanya hotspot di aera shaft bearing 2 (DE Motor) kemungkinan shaft bending

V. Rekomendasi :

- Perencanaan penggantian bearing motor
- ~~Pengecekan kelurusan shaft motor~~
- Pengecekan kondisi sealing bearing 2

JERANJANG, 13 MEI 2020

MENGETAHUI



RICKY RINALDI
SPS RSO

DISUSUN OLEH



HERMAWAN
PS PdM